

TeamLeader Integration

Connects **TeamLeader Focus** (calendar, tasks, contacts) with the Newo AI agent so callers and chat visitors can ask about availability, book service visits, reschedule, or cancel — entirely by voice or chat, no receptionist required.

What the AI can do

- **Check live availability.** Open time slots are pulled live from your technicians' TeamLeader calendars.
- **Book a service visit.** Once the customer picks a slot and gives their details, the agent creates the meeting in TeamLeader (and, optionally, a matching task) — the technician sees it on the calendar immediately.
- **Recognise returning customers.** When the customer mentions a phone number or email, the agent finds the matching TeamLeader contact and pre-loads their upcoming visits — works even **across sessions** (a customer who books today and calls tomorrow is recognised on the next call).
- **Cancel a visit, even one made on a different day.** The agent looks up the customer's upcoming visits in TeamLeader, presents them, and cancels the one the customer points to.
- **Reschedule a visit, even from a different session.** Same lookup; the agent moves the meeting to the new time and keeps the original duration. If the requested slot is taken it reads back the next 3 days that have openings.
- **Route to the right technician.** With one default technician the agent always books to that person. With multiple technicians the agent picks the right one for each request — by job title, by team, or by an operator-defined rule (see *Multi-technician routing* below).
- **Handle complaints gracefully.** A customer calling to complain is never offered a tool — the agent apologises, takes a message, and lets the operator know there is no booking change to make.

Features at a glance

Feature	Included
Voice + chat (WhatsApp, web, SMS, Telegram, phone)	✓
Live availability lookup	✓
Book a service visit	✓
Cancel a service visit (current session and across sessions)	✓
Reschedule a service visit (current session and across sessions)	✓
Recognise returning customers by phone or email	✓
Disambiguate when the customer has multiple visits — agent asks which one	✓
Auto-create the contact in TeamLeader during booking	✓
Reuse an existing TeamLeader contact when phone matches	✓
Pick the right technician based on the request	✓ (4 routing modes)
Create a TeamLeader task alongside each meeting	✓ (optional)
OAuth 2.0 with automatic token refresh	✓

How the customer hears the slots

The agent always reads slots back as **time ranges**, e.g. "10 AM to 11 AM", so the caller hears both the start time and the duration. If the customer asks for a specific time the agent confirms with a single sentence ("Yes, May 7 from 10 AM to 11 AM is available"); if it isn't available the agent says so and reads the next three days that have openings.

Multi-technician routing

If your TeamLeader account has more than one bookable technician, you choose **how the agent decides which one to use** by changing the **Employee Selection Type** setting:

Mode	When to use it	What the agent matches against
single (default)	One technician handles everything.	Always books to the <i>Default Technician</i> you select in Newo.
by_function	Each technician has a clear job title in TeamLeader (HVAC Technician, Plumber, Electrician).	The technician's Function field on their TeamLeader profile.
by_team_name	You organise technicians into TeamLeader Teams (HVAC Field Ops, Plumbing Crew).	The team name the technician belongs to.
by_config	You want to write the rules yourself in plain text.	Free-text routing rules you paste into Newo (one rule per line).

The agent reads the customer's request and picks the technician(s) whose function / team / rule matches best. The booking is then placed on **that** technician's calendar — not on the static default.

Setup

1. Prepare your TeamLeader account

Most of the work is one-time configuration on the **TeamLeader** side. The Newo side is just pasting credentials.

1a. Confirm at least one user can take bookings

Open **TeamLeader → Settings → Users**. Make sure every technician you want the agent to book is **active** and has:

- a working email address (used for the dropdown in Newo)
- a time zone that matches where they actually work (controls the slot times)

If you only have one user, that's fine — leave **Employee Selection Type** at `single` and skip the rest of step 1.

1b. Set the Function for each technician (only needed for `by_function` routing)

Open the technician's profile in TeamLeader and fill in the **Function** field with their job title — for example `HVAC Technician`, `Plumber`, `Electrician`, `Roof Specialist`. This is the field the agent matches against when **Employee Selection Type** is `by_function`.

You can edit it at any time; click **Publish All** in Newo afterwards so the new function is reflected in the agent.

1c. Create teams (only needed for `by_team_name` routing)

Open **TeamLeader** → **Settings** → **Teams**.

1. **Create a team** for each pool of work — *HVAC Field Ops, Plumbing Crew, Roof Specialists*, etc. The name you type here is exactly what the agent matches against, so make it descriptive.
2. **Add members**. Open the team and add every user that should belong to it. A user can be in multiple teams (e.g. a technician who covers both *HVAC* and *Heating Maintenance*).
3. **Pick a team lead**. TeamLeader requires every team to have a lead — pick whichever user is the most senior member.

After every team change in TeamLeader, click **Publish All** in Newo so the new team list is fetched into the agent's settings.

1d. Configure working hours

The agent only offers slots when the technician is normally working. TeamLeader exposes this through the user's **availability schedule**. To configure it:

1. Open **TeamLeader** → **Calendar**.
2. Click the **gear icon** in the top-right of the calendar view → **Working hours** (or in some locales **Availability**).
3. Set the working pattern per technician — typically *Mon–Fri 09:00–17:00* with a lunch break.
4. Save.

The agent reads this schedule on every availability check, so any change in TeamLeader takes effect on the very next customer turn — no Newo republish needed.

1e. Decide what to create on each booking

The integration can create **either** a meeting **or** a task **or both** for every booking. The recommended default is *both* — the meeting blocks the technician's calendar slot and the task gives the office a record they can run reports against. You'll select this in step 3 (`teamleader_booking_resource_type`).

2. Get your TeamLeader credentials

The integration uses **OAuth 2.0**, so you'll register a private application in TeamLeader's Marketplace.

1. Go to <https://marketplace.focus.teamleader.eu/integrations/private> and sign in with your TeamLeader account.
2. Click **Create integration**.
3. Pick a name (e.g. *Newo AI Agent*) and an icon.
4. **Redirect URI** — paste the value Newo gives you in the `teamleader_oauth_code` attribute description (it's pre-generated for your project).
5. Pick the **scopes** the agent will use:
 - `users` (read users + teams)
 - `events` (create / list / update / delete meetings)
 - `todos` (create / list / update / delete tasks — only if you'll use `task` or `both` mode)
 - `contacts` (look up and create the customer record)
 - `companies` (optional, only if you'll route inbound calls to companies)
6. Save the integration. TeamLeader shows you the **Client ID** and **Client Secret** — copy both.

3. Configure in Newo

1. Open the project and find the **Module - TeamLeader Integration** attribute group.
2. Paste the **Client ID** into `teamleader_client_id` and **Client Secret** into `teamleader_client_secret`.
3. (Optional but recommended) review the feature toggles:

Setting	Purpose	Default
Availability Feature	Live slot lookup on/off	on
Booking Feature	Booking creation on/off	on
Cancellation Feature	Cancellation on/off	on
Reschedule Feature	Reschedule on/off	on
Existing Client Lookup	Auto-recognise returning customer by phone, pre-load their visits	on
Auto-Setup Canvas Scenarios	Publishes the booking / cancel / reschedule scenarios on first install	on
Booking Resource Type	What to create per booking — meeting, task, or both	both
Default Country	Two-letter country code used when the customer doesn't provide one	BE
Employee Selection Type	How to pick the technician — single / by_function / by_team_name / by_config	single
Default Technician	Which TeamLeader user to book by default (used when Selection Type is single)	<i>(picked from a dropdown after first publish)</i>
Custom Employee Mapping	Free-text routing rules (only used with by_config)	<i>(empty)</i>
Default Meeting Duration (minutes)	Slot length when the customer doesn't specify	60
Availability Lookahead (days)	How far ahead the agent fetches the calendar	14

4. Click **Publish All**. The integration sets up the technical plumbing and produces an **install link** in the description of the `teamleader_oauth_code` attribute.
5. **Click the install link**. TeamLeader will ask you to allow the integration. Approve, and TeamLeader will redirect you back with a code in the URL bar.
6. Copy that code into `teamleader_oauth_code` and click **Publish All** again. The agent will:
 - Exchange the code for tokens.
 - Pull the active user list and the team list.
 - Populate the **Default Technician** dropdown with every active TeamLeader user.
 - Add the booking / cancel / reschedule scenarios to the canvas.
7. Pick the **Default Technician** from the dropdown and click **Publish All** one final time.

You're live. Place a test chat or call to confirm.

4. Routing rules for `by_config` (only if Employee Selection Type is `by_config`)

In the `teamLeader_by_config_mapping` attribute, paste one rule per line in this format:

```
<email>:<full name>:<comma-separated keywords>
```

Example:

```
alice@yourco.com:Alice Cooper:HVAC, AC repair, heating, electrical  
bob@yourco.com:Bob Builder:Plumbing, sink leaks, pipe repair, drain cleaning
```

The agent reads the **conversation** alongside this list and picks up to 3 matching technicians. The exact emails and names of your active users are auto-listed at the bottom of the attribute description on every publish — copy them from there so you don't have to look them up.

You don't need to type user IDs anywhere — the agent resolves them automatically from the live user list.

Returning-customer recognition

When the customer mentions a phone number or email (and at least once on phone-channel calls), the agent:

1. Searches TeamLeader contacts for a match (E.164 phone number or email).
2. Loads the contact's profile silently in the background.
3. Pulls every upcoming meeting attached to that contact.

This means by the time the customer says *"please cancel my appointment"* the agent already has the visit in front of it — no awkward *"what's your booking ID?"* question. It also works **across sessions**: a customer who books today and calls tomorrow is recognised on the second call.

If the customer has more than one upcoming visit the agent reads the list back and asks which one — never guesses.

Customizing the agent

The booking / cancel / reschedule scenarios on canvas are fully editable. Common tweaks:

- **Ask for more info during booking.** Add steps like "ask for the gate code" or "ask if pets are at home".
- **Change the consultation tone.** The wording of every scenario can be rewritten as long as the **code-phrases** and **scenario stages** below stay intact.
- **Restore defaults.** If you break a scenario, delete it from the canvas and click **Publish All** — the original is restored from the integration's library.

Prompt sections — what the agent reads, what the integration writes

The integration speaks to the canvas through five **prompt sections**. Each one is a slot on the canvas the integration writes into in the background; the scenarios read those slots and react to what's there. **Do not rename these sections, do not delete them, and do not write to them by hand.** If you remove or rename a section, the agent loses the connection to TeamLeader and the corresponding scenario silently breaks.

Section	Written by	What it carries	Where it's read
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<AvailableSlots>	Availability flow after /events.list returns	The list of free service slots for the next few days, plus a status (In progress, success, error, awaiting)	Booking scenario step S3, Reschedule step R3
<BookingResult>	Booking flow after /meetings.schedule returns	Status of the most recent booking attempt (success, awaiting: city, error: ...) and the resulting meeting id / date / time	Booking scenario step S10
<CancellationResult>	Cancellation flow after /meetings.delete returns	Status of the most recent cancel (success, error: ...)	Cancellation scenario step C3
<RescheduleResult>	Reschedule flow after /meetings.update returns	Status of the most recent reschedule (success, error: ...) and the new date/time	Reschedule scenario step R6
<ExistingCustomerInfo>	Existing-client lookup after /contacts.list returns	Whether the caller was recognised and a short summary	Greeting / consultation flows, optional acknowledgement at conversation start

You'll see these section names hardcoded in the scenario bodies on the canvas. They are **not** something you change — they're the contract between the scenario and the integration's TeamLeader calls.

Scenario stages — keep the order and the code-phrases

Each of the three scenarios is broken into ordered **stages** (steps). The scenario text is editable, but the **stage order** and the **code-phrases** are required for the integration to fire correctly. Editing the wording around them is fine; editing or removing the stages themselves breaks the flow.

Booking — stages S1 → S12

Stage	What happens	Why it can't be removed
S1	Ask the customer to describe the service	Without a service description the agent can't title the booking.
S2	Ask for brand and model (e.g. <i>Carrier 24ABC6</i>)	Goes into meeting description so the technician arrives prepared.
S3	Code-phrase: <i>"Let me check the technician's calendar for you. Please give me a moment."</i>	This phrase fires the availability lookup. Without it <AvailableSlots> stays empty and the booking will never load slots.

S4	Customer picks a slot from the offered list	Anchors the booking to a real free slot.
S5	First and last name	Required by TeamLeader's contact creation.
S6	Phone in E.164 (+44..., +1...)	Required by TeamLeader. Domestic formats are rejected.
S7	Email	Optional in TL but improves the contact record.
S8	Address — street, city, postal code (all three)	TeamLeader rejects bookings with a missing postal code or city.
S9	Read the full summary back, ask "Can I go ahead and book?"	Customer confirmation gate.
S10	Code-phrase: <i>"I'm booking your service visit now. Please give me a moment."</i>	This phrase fires the booking. Without it <code>meetings.schedule</code> is never called.
S11	Tell the customer what to expect (technician will arrive in window, keep phone available).	UX.
S12	Hand off to Finish Conversation .	Closes the call gracefully.

Cancellation — stages C1 → C5

Stage	What happens	Why it can't be removed
C1	Code-phrase: <i>"Let me look up your appointment. Please give me a moment."</i>	This phrase fires the meeting lookup. Without it the agent has no list to cancel from.
C2	Reconfirm the target appointment day/time, ask "Are you sure?"	Confirmation gate before the destructive action.
C3	Code-phrase: <i>"Give me a moment, I'll cancel your appointment now."</i>	This phrase fires the cancel. Without it <code>meetings.delete</code> is never called.
C4	Ask if anything else (e.g. <i>"would you like to rebook?"</i>)	UX.
C5	Hand off to Finish Conversation .	Closes gracefully.

Reschedule — stages R1 → R7

Stage	What happens	Why it can't be removed
R1	Code-phrase: <i>"Let me look up your appointment. Please give me a moment."</i>	Fires the meeting lookup. Same phrase as C1 — same purpose.
R2	Confirm the target appointment, ask for the new day and time	Anchors which booking is being moved.

R3	Code-phrase: <i>"Let me check the technician's calendar for you. Please give me a moment."</i>	Fires availability for the new slot. Same phrase as S3.
R4	Customer picks a slot from the offered list	Anchors the reschedule to a real free slot.
R5	Read the reschedule summary back, ask "Can I go ahead?"	Customer confirmation gate.
R6	Code-phrase: <i>"Let me move that appointment to your new time. Please give me a moment."</i>	Fires the reschedule. Without it <code>meetings.update</code> is never called.
R7	Hand off to Finish Conversation .	Closes gracefully.

Editing rules — what's safe and what isn't

✅ Safe to edit	❌ Don't touch
Wording around the code-phrases (e.g. <i>"Sure, let me check that for you. Let me check the technician's calendar for you. Please give me a moment."</i>)	The exact code-phrase text — even a punctuation or capitalisation difference will stop the lookup
Adding new questions (e.g. ask for gate code, pet info, allergy notes) between existing stages	Removing or reordering stages
Rewording the customer-facing summary in S9 / R5	The placeholders for <code><AvailableSlots></code> / <code><BookingResult></code> / <code><CancellationResult></code> / <code><RescheduleResult></code> / <code><ExistingCustomerInfo></code>
Adding a new branch ("if the customer is a VIP, do X")	Renaming or deleting the prompt sections from the scenario
Translating to another language	Translating the code-phrases themselves — the integration matches them in English. Keep the English phrase, even in a multilingual setup. The agent will speak the rest of the scenario in the customer's language naturally.

If you accidentally break a scenario, the safest recovery is to **delete it from the canvas** and click **Publish All** — the integration's library re-installs the shipped version.

Frequently asked questions

Q. Can the customer reschedule in one step? Yes — the agent moves the existing meeting to the new time without cancelling and re-booking. The original meeting ID stays the same so reports stay consistent.

Q. We just promoted Alice from HVAC to Plumbing. When does the agent see it? After the next **Publish All**, the agent re-fetches the user list and the team list from TeamLeader. The new function or team membership takes effect on the very next customer turn.

Q. Does the agent answer policy questions ("how soon can a technician arrive?", "do you charge a call-out fee?")? Yes — those come from the **Business Context** on the canvas, not from TeamLeader.

Q. What happens if the customer asks for a slot that's not available? The agent says so plainly and reads the next 3 days that have openings, so the customer can pick another time without going round in circles.

Q. The agent is booking on the wrong technician. Three things to check:

1. **Employee Selection Type** matches your setup (`by_function` if you use Function fields; `by_team_name` if you use Teams; `by_config` if you wrote routing rules).
2. The technician's **Function / Team** / mapping line is filled in correctly in TeamLeader and contains the keyword the customer is using (e.g. `Plumbing` if the customer says *"sink leak"*).
3. You clicked **Publish All** in Newo after the change in TeamLeader.

Q. The customer says "I want to file a complaint" — what happens? The agent recognises that this is not a booking task, apologises, takes a message (name, phone, callback time, what the complaint is about), and lets the customer know a human will follow up. No tool is fired — your TeamLeader stays untouched.

Q. Two customers try to book the same slot. TeamLeader serializes the bookings. The second request comes back with a *"slot no longer available"* error; the agent apologises and offers the remaining slots from the next three days that have openings.

Q. The OAuth code expired before I could paste it. Click the install link in the `teamleader_oauth_code` attribute description again — you get a fresh code every time.

Q. What happens if the access token expires? The integration auto-refreshes on 401 using the refresh token — no manual action needed. If the refresh token itself becomes invalid (e.g. the customer revoked the app in TeamLeader), the agent surfaces a clear *"re-authorize required"* state in the install status. Re-paste a new OAuth code and Publish All.

Q. Can I run multiple TeamLeader companies on one Newo project? One TeamLeader account per integration instance. To serve a second company, deploy a second Newo project.

Limitations

- **Reschedule preserves the original meeting duration** unless the customer explicitly asks for a different one (e.g. *"move it to Friday at 2 PM and make it 90 minutes"*).
- **Tasks-only mode** (`teamleader_booking_resource_type=task`) is supported but the search-for-existing-visit feature only knows about meetings, so a task-only booking made in one session may not be found when the customer calls back on a different day. If you rely on cross-session cancellation/reschedule, leave the setting at `both` (the default).
- **One TeamLeader account per Newo project.**
- **Working hours** (configured per user in TeamLeader) are read live on every availability check; **break times** during a working day are honoured only if the technician's calendar already has an event blocking that time.

Multi-technician routing — choosing the right mode

Quick decision tree:

- *We only have one technician, or one person handles everything.* → `single` .
- *Each technician has a clear job title (HVAC Tech, Plumber, Electrician).* → `by_function` . Lowest setup cost — you only fill in the Function field on each user.
- *We organise people into Teams in TeamLeader (HVAC Field Ops, Plumbing Crew).* → `by_team_name` . Best when teams already exist for other reasons (rota, reporting).
- *We want bespoke rules — Alice does HVAC and after-hours emergencies, Bob does plumbing on weekdays only.* → `by_config` . Most flexible, also the most prose to write.

You can change the mode at any time, click **Publish All**, and the agent reflects the new behaviour on the next customer turn.

All settings reference

Every attribute the integration adds to the **Module - TeamLeader Integration** group in Newo. Operators typically only touch the "Primary" rows; the rest are good out of the box.

Primary (set these during setup)

Attribute	Title in UI	Purpose	Default
teamleader_client_id	Application ID	OAuth Client ID from TeamLeader Marketplace.	<i>(empty)</i>
teamleader_client_secret	Application Secret	OAuth Client Secret from TeamLeader Marketplace.	<i>(empty)</i>
teamleader_oauth_code	OAuth Code	One-time authorization code returned by TeamLeader after the operator clicks the install link.	<i>(empty)</i>
teamleader_assignee	Default Technician	Dropdown of active TeamLeader users; the one to book by default in single mode and the source of the working-hours template in all modes.	<i>(picked after first publish)</i>
teamleader_employee_selection_type	Employee Selection Type	single / by_function / by_team_name / by_config.	single
teamleader_by_config_mapping	Custom Employee Mapping	Free-text routing rules (one per line). Used only with by_config.	<i>(empty)</i>
teamleader_booking_resource_type	Booking Resource Type	What to create per booking — meeting, task, or both.	both
teamleader_default_country	Default Country	Two-letter country code (ISO 3166-1) used on bookings when the customer doesn't say.	BE

teamleader_enable_check_availability	Availability Feature	Master on/off for the live slot lookup.	on
teamleader_enable_booking	Booking Feature	Master on/off for booking creation.	on
teamleader_enable_cancellation	Cancellation Feature	Master on/off for cancellation.	on
teamleader_enable_reschedule	Reschedule Feature	Master on/off for reschedule.	on
teamleader_enable_search_meetings	Existing Client Lookup	Master on/off for the auto-load-visits-by-phone behaviour at conversation start.	on
teamleader_default_meeting_duration_minutes	Default Meeting Duration (minutes)	Slot length when the customer doesn't specify.	60
teamleader_availability_lookahead_days	Availability Lookahead (days)	Maximum number of days ahead the agent fetches the calendar. The agent voices the first 3 days with openings.	14

Advanced fine-tuning (hidden by default)

There are advanced settings — labelled *Advanced* — *for admins only* in the UI — that let you fine-tune **what the agent extracts** from the conversation (which date, which time, which duration, which technician). They're hidden by default. Clear a value and **Publish All** to restore the shipped default.

Internal settings (OAuth tokens, the resolved TeamLeader endpoint, the live employees and teams cache, the slot map for the active call) are managed automatically and never need manual editing.

Troubleshooting

Symptom	Likely cause	Fix
Default Technician dropdown is empty after pasting OAuth code.	OAuth exchange failed — usually a wrong Client ID/Secret or the redirect URI doesn't match the one in TeamLeader Marketplace.	Re-check Client ID/Secret. In TeamLeader Marketplace open your private integration → Settings → confirm the Redirect URI matches what Newo gave you. Re-publish.
Default Technician dropdown only shows one user but you have	TeamLeader users marked inactive are filtered out at sync time.	In TeamLeader → Settings → Users, set the missing users to active. Publish All.

several active TeamLeader users.		
Booking goes on the wrong technician in multi-tech mode.	Function / Team / Mapping not configured to match the customer's wording, or Employee Selection Type still on single.	See FAQ "the agent is booking on the wrong technician".
Agent says "the scheduling system is temporarily unavailable" mid-conversation.	OAuth refresh failed — usually because the refresh token was rotated outside Newo (e.g. someone else used the same private app), or the integration was uninstalled in TeamLeader.	Re-paste a new teamleader_oauth_code and Publish All .
Customer's date/time in TeamLeader looks one day off for tasks.	TeamLeader stores the task's <i>due_on</i> in account-side timezone — the integration already compensates for negative-offset assignee timezones. If you still see a one-day drift, check that the technician's TeamLeader profile has the right time zone set.	TeamLeader → Settings → Users → open the technician → set Time zone. Publish All .
Cancel / reschedule doesn't find the visit when the customer calls a day later.	Either the Existing Client Lookup master switch is off, or the customer is using a phone number that wasn't on the original booking.	Turn on Existing Client Lookup . Ask the customer for the phone number or email used on the original booking.
Reschedule moves the meeting but task date stays old for a both booking made before the integration was upgraded.	Bookings made before v2.x don't carry the task-link marker.	Cancel + rebook once to get the new format; from then on reschedule moves both.
Two customers booked the same slot within a few seconds.	TeamLeader serialises but the second arrival hits a "slot no longer available" error.	Expected — agent apologises and reads the next 3 days with openings.

If you're stuck, contact your Newo onboarding rep with:

1. The text of the most recent agent message.
2. A screenshot of the **teamleader_oauth_code** **attribute description** (which shows the install_status).
3. A screenshot of the **TeamLeader Settings → Users** page (shows active users, function fields, time zones).